

YZV202E Optimization for Data Science Project Proposal

Inventory-Constrained Dynamic Bidding for One-Shot Auctions

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I. PROJECT DESCRIPTION

The idea of the project is the optimization of a model in which our business, with minimal financial expenditure and satisfying our needs, manages to purchase the goods it needs at that moment. We have limited the project to the level of one country within whose borders multiple companies are trying to purchase the goods they need at an auction held daily, which means that the companies are competitors to each other. At the auction venue, the rule is as follows - each party has the right to offer only one price, and the one who offers the highest wins the product. This means that there are no multiple rounds in which offers can be increased. It is a one-shot auction. Four different types of products are sold at the auction: milk, cream and dairy products, eggs, poultry, and beef. In order to limit the complexity of the project itself and to make modeling the solution easier, we will assume that our company knows at all times what the other companies have at their disposal, namely how many products they have in their inventory. Given that the auction is held daily and that every company participates in the auction every day, it is quite expected that sometimes a company will have no need for any product at that moment because its current inventory are sufficient, while it can happen that a company needs more than one product. Taking into account variations in market prices, as well as the inventory levels of individual products in our company and in other companies, the project aims to create a model that will enable our company to secure the necessary goods with minimal financial expenditure, that is, to maximize its profit.

II. PROBLEM DEFINITION

The objective of the project is to develop a model that minimizes budget spent over time while keeping the inventory above a minimum. Formally, the problem is defined as follows: Let

- $x_i^{(t)} \in \{0, 1\}$: binary decision variable indicating whether we bid for product i at time t ,
- $b_i^{(t)}$: auction cost of product i at time t ,

- $s_i^{(t)}$: available inventory of product i at time t ,
- α_i : minimum required inventory level for product i .

$$\min_x \sum_{t=1}^T \sum_{i=1}^N b_i^{(t)} x_i^{(t)} \quad s.t. \quad (1)$$

$$s_i^{(t)} \geq \alpha_i, \quad \forall i, \forall t$$

$$s_i^{(t+1)} = s_i^{(t)} + q_i^{(t)} x_i^{(t)} - d_i^{(t)} \quad (2)$$

where

- $q_i^{(t)}$ is the amount of product i being sold at time t ,
- $d_i^{(t)}$ is the depletion of product i at time t .

Since the competitors' inventory is dynamic, the market price for a given auction with product i will be dynamic as well. Simply said, if the inventory of a certain good at other companies are low, it means that the price of that product on the market will be higher, and vice-versa. Another complexity is that the quantity of each product i at a given auction round may not be the same at another round. This means that the bidding rule will have to optimize between minimizing cost spent per unit product whilst satisfying the inventory constraint. Since the bidding prices of competing companies at time t are unknown, the model should be able to predict them using least squares linear regression. For each product i , the prediction is based on the quantity of that product offered at the auction and the amount of that product currently held in the competitors' inventory.

III. DATASET

To create the auction system, we will be using slaughtering statistics in slaughterhouses [1], slaughtering statistics outside slaughterhouses [2], poultry production data [3], and cows' milk collection and dairy production data [4] from EURO-STAT. These datasets provide monthly production statistics across European countries.

Specifically, the slaughtering datasets contain monthly meat production volumes, including bovine, pig, and sheep (mutton and lamb) meat, reported for each country from September

2009 to February 2026. The poultry production dataset includes monthly statistics on chicken meat and egg production, covering the period from January 1967 to February 2026. The milk dataset provides monthly data on raw cow milk delivered to dairies, as well as processed dairy products such as cheese and butter, for each country from January 1968 to February 2026.

Together, these datasets cover the primary components of livestock-based food production and provide a suitable basis for modeling productions over time. This makes them well-suited for simulating and evaluating auction-based mechanisms in our case.

IV. METHODOLOGY

We will first create the auction simulation in Python utilizing data from the datasets described in III. Then the problem will be separated into 3 problems with increasing complexity, and we will solve from easiest to hardest sequentially.

A. Deterministic Bidding Setup

To create an optimal baseline, we initially assume an offline environment, where we know the environment completely beforehand. Specifically, we assume that the exact quantities of the auctioned items $q_i^{(t)}$ and their respective bidding cost to win $b_i^{(t)}$ are deterministic and known to us across the entire time period T . By removing the stochasticity from the environment, the objective simplifies to cost minimization. The problem reduces to a standard Integer Linear Programming (ILP) problem, where the action becomes deciding at which discrete timestamps bidding $x_i^{(t)} \in \{0, 1\}$ minimizes the total cumulative cost over the timesteps.

$$\min \sum_i^N b_i \quad \text{s.t.} \quad \alpha_i \leq q_i \quad (3)$$

where α_i is the minimum amount of product quantity q_i that we should have after the round ends, subscript i indicates the index of the product of interest, and N is the number of different products that is auctioned, e.g. milk, eggs, etc.

B. All-Seeing Bandit-Style Problem

Every buyers' budget is unlimited, and at each auction, their bidding amount is completely known to us (hence the name All-Seeing). Other buyer's bidding amount dynamically change over time and our goal is to optimize budget spent over time T while satisfying the inventory constraints.

A naive approach would be to always buy product i if $q_i < \alpha_i$, but this ignores the dynamic nature of the auction's market price. Another naive approach would be to always buy product i when the cost is below some predetermined threshold, but again such approach would not minimize total budget spent and may even violate the inventory constraint.

Our approach is to utilize the Multi-Armed-Bandit (MAB) algorithm [5] while adapting a form of Lyapunov Optimization [6] so that it provides a way to optimize between cost minimization and inventory safety. The full formal analysis

of our version is written in the Appendix. For each product i , we define a deficit state,

$$H_i(t) = \beta_i - s_i^{(t)} \quad (4)$$

$$\beta_i > \alpha_i \quad (5)$$

where β_i is the soft inventory constraint to ensure the hard constraint α_i is always satisfied. The standard Lyapunov analysis becomes,

$$\min_{x_i^{(t)}} \sum_{i=1}^N \left(V b_i^{(t)} - H_i(t) q_i^{(t)} \right) x_i^{(t)} \quad (6)$$

Accordingly, the bidding rule becomes,

$$x_i^{(t)} = \begin{cases} 1, & \frac{b_i^{(t)}}{q_i^{(t)}} < \frac{H_i(t)}{V} \\ 0, & \text{otherwise} \end{cases} \quad (7)$$

Here, V is a hyperparameter that controls the importance between cost minimization and inventory safety. If V is high, then more emphasis on cost minimization is given, and vice versa. This bidding rule will prioritize buying product i if it is considerably cheap while maintaining the hard constraint in (1).

C. Partially Blind Bandit-Style Problem

The problem formulation is the same as IV-B, but now the bidding amount of other buyers is not known to us, but in turn, their respective inventory is known to us (hence the name Partially Blind). We can assume that the other buyers bid linearly w.r.t. to their inventory conditions (and we will model the environment as such). Thus, a trivial solution is to approximate the market price (highest bid among other buyers) for an auction selling product i as,

$$\mathbb{E}[C_{i,o}|K_o] = K_o^T \theta_{i,o} \quad (8)$$

where,

- C is the cost distribution for buyer o and product i .
- K is the context distribution, in this case the buyer o 's inventory condition.
- θ is the learnable parameter vector for pair product i and buyer o .

The market price for product i at time t is then

$$P_i^{(t)} = \max_o \mathbb{E}[C_{i,o}|K_o] \quad (9)$$

and our bid becomes

$$b_i^{(t)} = P_i^{(t)} + \epsilon \quad (10)$$

where ϵ is the smallest number that guarantees win. Here, we can get data and fit this linear regression model online using incremental Ordinary Least Squares (OLS).

REFERENCES

- [1] EUROSTAT, “Slaughtering in slaughterhouses (apro_mt_pwgtn)”
- [2] EUROSTAT, “Slaughtering outside slaughterhouses (apro_mt_pslthm)”
- [3] EUROSTAT, “Poultry production (apro_ec_poulm)”
- [4] EUROSTAT, “Milk collection and dairy products (apro_mk_colm)”
- [5] P. Auer, N. Cesa-Bianchi, and P. Fischer, “Finite-time analysis of the multiarmed bandit problem,” *Machine Learning*, vol. 47, pp. 235–256, 2002.
- [6] M. J. Neely, *Stochastic Network Optimization with Application to Communication and Queueing Systems*. San Rafael, CA, USA: Morgan & Claypool, 2010.

APPENDIX

Optimality Analysis

We want to satisfy the hard constraints $s_i^{(t)} > \alpha_i \forall_{t,i}$ while minimizing total cost over time. Thus, we need to define a state function $H(t)$ that gives the difference of current s and its respective α . Given the inventory dynamics

$$s_i(t+1) = s_i(t) + q_i(t)x_i(t) - d_i(t),$$

we can define the deficit state function w.r.t. hyperparameter β_i (an analysis of this parameter will be provided shortly)

$$H_i(t) = \beta_i - s_i(t),$$

$$\beta_i > \alpha_i.$$

Then, by definition,

$$H_i(t+1) = H_i(t) - q_i(t)x_i(t) + d_i(t).$$

By this we can define the Lyapunov function

$$L(H(t)) = \frac{1}{2} \sum_{i=1}^N H_i(t)^2,$$

Since our current decision $x_i(t)$ depends on $H_i(t)$, we can define the one-step conditional drift

$$\Delta(H(t)) = \mathbb{E}[L(H(t+1)) - L(H(t)) \mid H(t)].$$

In order to satisfy the constraint, an equivalent goal is to minimize $\Delta(H(t))$ with β big enough that the constraint is always satisfied.

To achieve the main goal of minimizing cost over time, we can define a new target

$$J(x(t)) = \mathbb{E}\left[\sum_{i=1}^N b_i(t)x_i(t) \mid H(t)\right]$$

Using these targets, we can create a new weighted target. We can assume $0 \leq q_i(t) \leq q_i^{\max}$ and $0 \leq d_i(t) \leq d_i^{\max}$ for all i, t . Then there exists a finite constant $B > 0$ such that

$$\begin{aligned} & \Delta(H(t)) + V J(x(t)) \\ & \leq B + \sum_{i=1}^N H_i(t) \mathbb{E}[d_i(t) \mid H(t)] \\ & + \mathbb{E}\left[\sum_{i=1}^N (Vb_i(t) - H_i(t)q_i(t))x_i(t) \mid H(t)\right]. \end{aligned}$$

Here $V > 0$ is the hyperparameter that determines the importance of minimizing $J(x(t))$ over satisfying the constraints, represented with $\Delta(H(t))$. We will provide the derivation shortly. Since only the last part of the right side of the inequality is under our control, minimizing it is equivalent to minimizing the left side

$$\min_{x(t)} [\Delta(H(t)) + V J(x(t))] \equiv \min_{x(t)} \sum_{i=1}^N (Vb_i(t) - H_i(t)q_i(t))x_i(t).$$

If $x_i(t) \in \{0, 1\}$ and the decisions are separable across i , then the optimal policy is

$$x_i(t) = \begin{cases} 1, & H_i(t)q_i(t) > Vb_i(t), \\ 0, & H_i(t)q_i(t) \leq Vb_i(t). \end{cases}$$

Equivalently, for $q_i(t) > 0$,

$$x_i(t) = 1 \iff \frac{b_i(t)}{q_i(t)} < \frac{H_i(t)}{V}.$$

Here, the bidding rule is to bid whenever the price of product is cheap and/or the deficit state is high.

Proof of Upper Bound

From

$$H_i(t+1) = H_i(t) - q_i(t)x_i(t) + d_i(t),$$

we have

$$\begin{aligned} (H_i(t+1))^2 &= (H_i(t))^2 + (d_i(t) - q_i(t)x_i(t))^2 \\ &\quad + 2H_i(t)(d_i(t) - q_i(t)x_i(t)), \end{aligned}$$

subtracting both sides by $(H_i(t))^2$ gives

$$\begin{aligned} (H_i(t+1))^2 - (H_i(t))^2 &= (d_i(t) - q_i(t)x_i(t))^2 \\ &\quad + 2H_i(t)(d_i(t) - q_i(t)x_i(t)). \end{aligned}$$

Summing over i , multiplying by $1/2$, and taking conditional expectation yields

$$\begin{aligned} \Delta(H(t)) &= \mathbb{E}\left[\frac{1}{2} \sum_{i=1}^N (d_i(t) - q_i(t)x_i(t))^2 \mid H(t)\right] \\ &\quad + \sum_{i=1}^N H_i(t) \mathbb{E}[d_i(t) - q_i(t)x_i(t) \mid H(t)]. \end{aligned}$$

By the assumption of bounds of $d_i(t)$ and $q_i(t)$, there exists $B < \infty$ such that

$$\mathbb{E}\left[\frac{1}{2} \sum_{i=1}^N (d_i(t) - q_i(t)x_i(t))^2 \mid H(t)\right] \leq B.$$

Hence

$$\Delta(H(t)) \leq B + \sum_{i=1}^N H_i(t) \mathbb{E}[d_i(t) \mid H(t)]$$

$$-\mathbb{E} \left[\sum_{i=1}^N H_i(t) q_i(t) x_i(t) \mid H(t) \right].$$

Adding $V J(x(t))$ to both sides gives

$$\begin{aligned} & \Delta(H(t)) + V \mathbb{E} \left[\sum_{i=1}^N b_i(t) x_i(t) \mid H(t) \right] \\ & \leq B + \sum_{i=1}^N H_i(t) \mathbb{E}[d_i(t) \mid H(t)] \\ & + \mathbb{E} \left[\sum_{i=1}^N (V b_i(t) - H_i(t) q_i(t)) x_i(t) \mid H(t) \right]. \end{aligned}$$

The first two terms on the right-hand side are independent of the control action, so minimizing the drift-plus-penalty upper bound reduces to minimizing

$$\sum_{i=1}^N (V b_i(t) - H_i(t) q_i(t)) x_i(t).$$

If decisions are separable and binary, the minimizer is obtained by choosing $x_i(t) = 1$ whenever

$$V b_i(t) - H_i(t) q_i(t) < 0,$$

which is equivalent to

$$H_i(t) q_i(t) > V b_i(t).$$

Analysis of β

As one can see, minimizing $\Delta(H(t))$ is equivalent to minimizing $L(H(t+1))$. L here is quadratic, meaning that minimizing this target can be seen geometrically as bounding q_i to always be in some radius around β_i . Thus, if $\beta_i - \alpha_i$ is not big enough, the constraint can be violated.

Since we know that the radius mentioned is determined by the supply s_i 's dynamic, a possible solution to this is to set

$$\beta_i = \alpha_i + d_i^{\max}$$

since d_i is assumed to be bounded, the supply dynamic is always $q_i(t) x_i(t) - d_i(t)$, and the lower bound over all time t of the dynamic is $-d_i^{\max}$.

A question may arise: "Why not buy every time the price is cheap?". Here, as in real-life, food has expiration and is included in d_i , so it is not a wise decision to overstock on food.